The documentation and process conversion measures necessary to comply with this amendment shall be completed by 7 December 2001.

INCH POUND

MIL-PRF-19500/427H
AMENDMENT 2
7 September 2001
SUPERSEDING
AMENDMENT 1
24 October 2000

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER,
TYPES 1N5614, 1N5616, 1N5618, 1N5620, 1N5622, 1N5614EG1, 1N5616EG1, 1N5618EG1, 1N5620EG1, 1N5622EG1, 1N5614UL, 1N5616UL, 1N5618UL, 1N5620UL, 1N5622UL, 1N5614US, 1N5616US, 1N5618US, 1N5620US, 1N5622US, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This amendment forms a part of MIL-PRF-19500/427H, dated 8 July 2000 and is approved for use by all Departments and Agencies of the Department of Defense.

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* Add the following paragraph:

"3.6. Marking. Marking shall be in accordance with MIL-PRF-19500."

* 3.6 Marking of UL and US version devices; delete "3.6" and substitute "3.6.1".

* Add the following paragraph:

"4.2.1 JANHC and JANKC die. Qualification shall be in accordance with appendix H of MIL-PRF-19500."

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* 4.3.1 Delete and substitute:

"4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: Method 1038 of MIL-STD-750, condition B, T_A = 50°C maximum, V_R = full rated V_{RWM} (see 1.3); f = 50-60 Hz, I_O = 1 A dc minimum, (see 4.5.7 and 4.5.7.1)."

* 4.3.2, delete and substitute:

"4.3.2 Screening (JANHC and JANKC). Screening of die shall be in accordance with appendix H of MIL-PRF-19500. As a minimum, die shall be 100-percent probed to ensure compliance with group A, subgroup 2. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements."

* 4.4, delete and substitute:

" 4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed in accordance with E.5.3.1d of MIL-PRf-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2)."

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* 4.4.1, add the following conditions:

"a. I_M measuring current - - - - - - - 10 mA.

b. I_H forward heating current - - - - - 10 A.

c. t_H heating time - - - - - - - - - - 10 ms.

d. t_{MD} measurement delay time - - - - - 100 µs maximum.

The maximum limit for $Z_{\theta JX}$ in group A, subgroup 2 is $Z_{\theta JX}$ (maximum) = 4.5°C/W."

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* 4.4.2.1, delete and substitute:

"4.4.2.1 Group B inspection, appendix E, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B4	1037	$I_O = I_O$ (rated) minimum; V_R = rated V_{RWM} (see 1.3).
B5	1027	$I_O = I_O$ (rated) dc minimum; T_A =100°C max. apply V_R = rated V_{RWM} (see 1.3 and 4.5.7); adjust T_A and I_O to achieve T_J = +275°C minimum; f = 50-60 Hz.
	or	$I_O = I_O$ (rated) dc minimum; T_A = +50°C max. apply V_R = rated V_{RWM} (see 1.3 and 4.5.7); adjust T_A and I_O to achieve T_J = +200°C minimum for 1,000 hours at double the sample size; f = 50-60 Hz.
B6	3101 or 4081	+25°C ≤ T_A ≤ +35°C (recorded before test is performed); $R_{\theta JL}$ (maximum) ≤ 38°C/W; L = .375 inch (9.53 mm) (see figure 6). For surface mount devices (UL and US version), $R_{\theta JEC}$ ≤ 7°C/W.
B7		Peak reverse power, see 4.5.4 and figure 7 herein. P_{RM} ≥ 500 W. Test shall be performed on each sublot; sampling plan n = 10, c = 0, end-points, see 4.4.2."

* 4.4.2.2, delete and substitute:

" 4.4.2.2 Group B inspection, Appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1027	T_A = +50° C max. adjust T_A and I_O to achieve the required T_J; f = 50-60 Hz; apply V_R = rated V_{RWM} (see 1.3, 4.5.7, and 4.5.7.1)."

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* 4.4.3.1, C6, method 1026, conditions column; delete and substitute:

"T_A = 50°C max, adjust T_A and I_O to achieve the required T_J; f = 50-60 Hz; apply V_R = rated V_{RWM} (see 1.3, 4.5.7, and 4.5.7.1)."

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* 4.4.4, delete and substitute:

"4.4.4 Group E inspection. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the associated specification that did not request the performance of group E inspection, the tests specified herein must be performed to maintain qualification. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IX of MIL-PRF-19500 and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein; except, $Z_{\theta JX}$ need not be performed. See table II for delta limits when applicable.

* 4.5.3, delete and substitute:

"4.5.3 Scope display evaluation. Scope display evaluation shall be sharp and stable in accordance with method 4023 of MIL-STD-750. Scope display may be performed on ATE (automatic test equipment) for screening only with the approval of the qualifying activity. Scope display in group A shall be performed on a scope."

* 4.5.7.1, delete and substitute:

"4.5.7.1 Mounting conditions. Any clips or heat sink mounting configurations may be utilized provided that I_O is adjusted such that the junction temperature of each diode is maintained at T_J = 135°C minimum for burn-in and T_J = 150°C minimum for life testing."

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TABLE I, subgroup 4, as last test; add "| Scope display | 4023 | See 4.5.3. n = 116, c = 0 |".

TABLE I, subgroup 6, conditions; delete "I_{FSM} = rated (see 1.3)" and substitute "I_{FSM} = 30 A (pk)".

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TABLE II, title; add "4/".

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Custodians:
Army - CR
Navy - NW
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

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